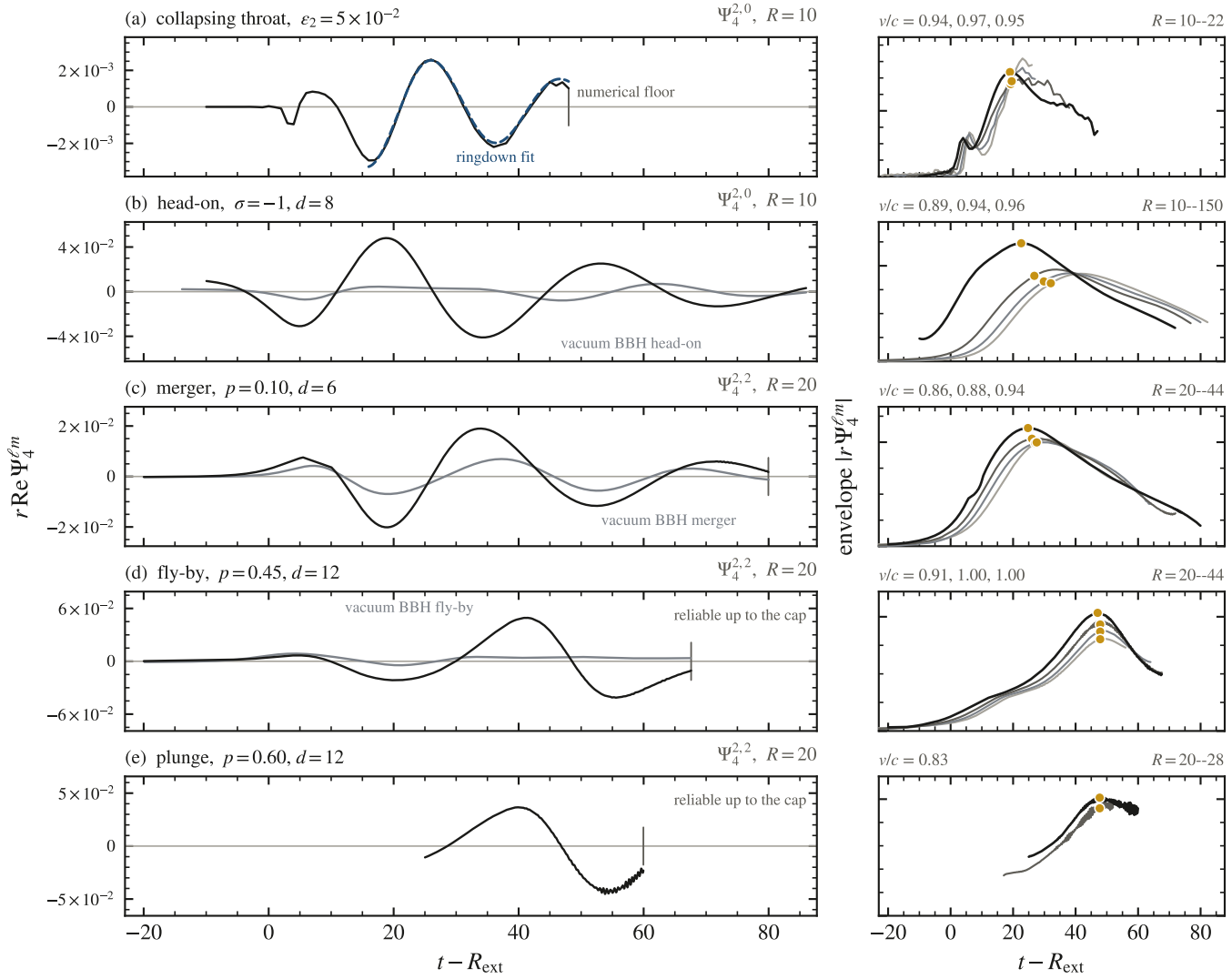

(a) collapsing throat, $\varepsilon_2 = 5\times10^{-2}$
$\Psi_4^{2,0}$, $R=10$
numerical floor
ringdown fit
v/c = 0.94, 0.97, 0.95
R = 10--22
(b) head-on, $\sigma=-1$, $d=8$
$\Psi_4^{2,0}$, $R=10$
vacuum BBH head-on
v/c = 0.89, 0.94, 0.96
R = 10--150
(c) merger, $p=0.10$, $d=6$
$\Psi_4^{2,2}$, $R=20$
vacuum BBH merger
v/c = 0.86, 0.88, 0.94
R = 20--44
(d) fly-by, $p=0.45$, $d=12$
$\Psi_4^{2,2}$, $R=20$
vacuum BBH fly-by
reliable up to the cap
v/c = 0.91, 1.00, 1.00
R = 20--44
(e) plunge, $p=0.60$, $d=12$
$\Psi_4^{2,2}$, $R=20$
reliable up to the cap
v/c = 0.83
R = 20--28
$r\,\mathrm{Re}\,\Psi_4^{\ell m}$
envelope $|r\,\Psi_4^{\ell m}|$
$t - R_{\rm ext}$
$t - R_{\rm ext}$